

Title: Urban Ecology and Arboriculture: A Study of Different Avenue Trees and Their Importance

Aim: To collect information on different Avenue trees, document their taxonomic classification, analyze their ecological, medicinal, and economic importance in urban environments, and arrive at a conclusion based on the analysis of data.

Project Writing

1. Name of the student and concerned details

- **Name:** Mayur Shrikrushna Mali
- **Standard:** 12th (Science)
- **Subject:** Biology
- **Project Topic:** 12 (Collect the information of different Avenue trees and their importance)
- **Academic Year:** 2026-2027

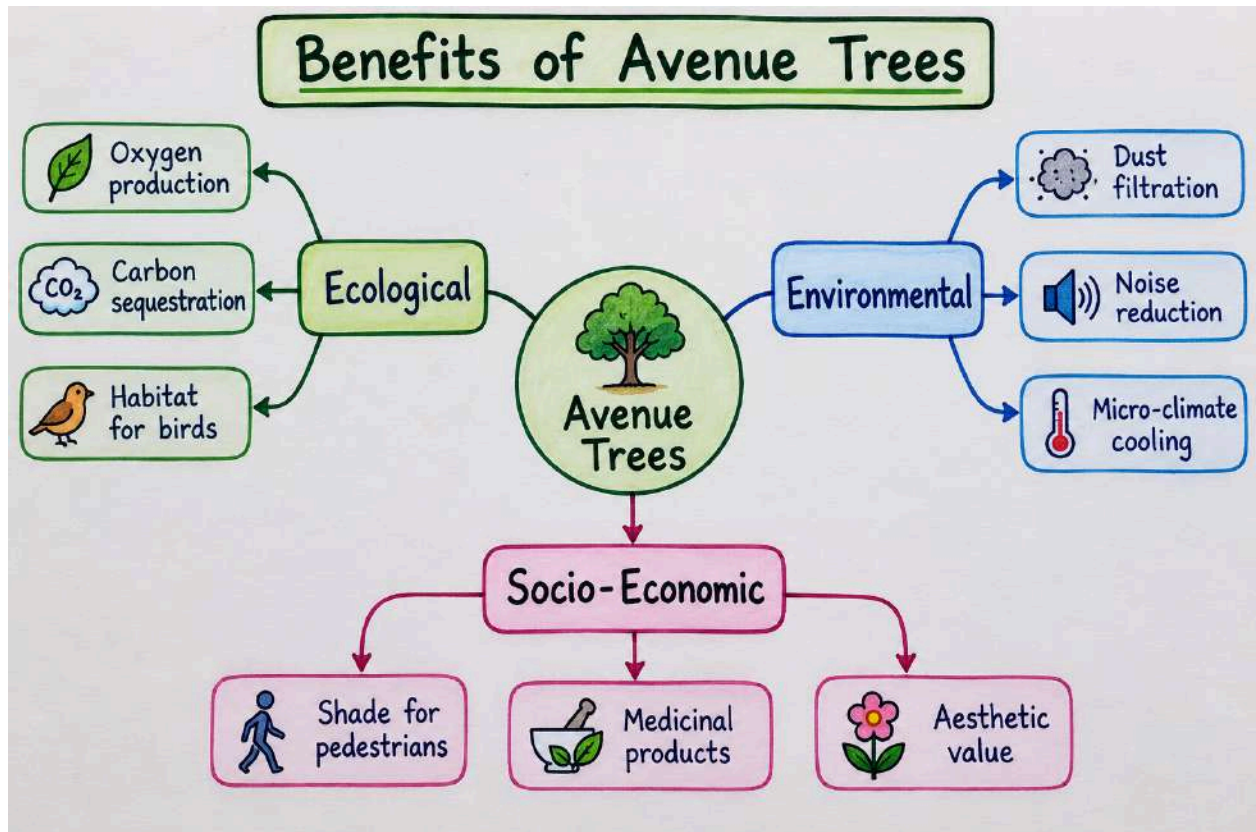
2. Acknowledgements:

I would like to express my sincere gratitude to my Biology teacher for their invaluable guidance in planning the approach and methodology for this project. Their continuous support helped me understand botanical and ecological concepts clearly. I am also deeply thankful to our Principal for providing the necessary academic environment. Lastly, I extend my heartfelt thanks to my parents and classmates for their encouragement and assistance in compiling this data.

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(Action: Draw a neat concept map titled "Benefits of Avenue Trees".)

- **Center Circle:** Avenue Trees
- **Branches pointing outwards:**
 - *Ecological:* Oxygen production, Carbon sequestration, Habitat for birds.
 - *Environmental:* Dust filtration, Noise reduction, Micro-climate cooling.
 - *Socio-Economic:* Shade for pedestrians, Medicinal products, Aesthetic value.



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3. Table of contents

1. Title and Aim
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4. Table of contents
5. Actual Report **a.** Introduction of the topic and need for project **b.**
6. Description of method/s used **c.** Data collected (Information on various Avenue Trees)
d. Analysis of collected data
e. Conclusion
7. Bibliography

4. Actual Report

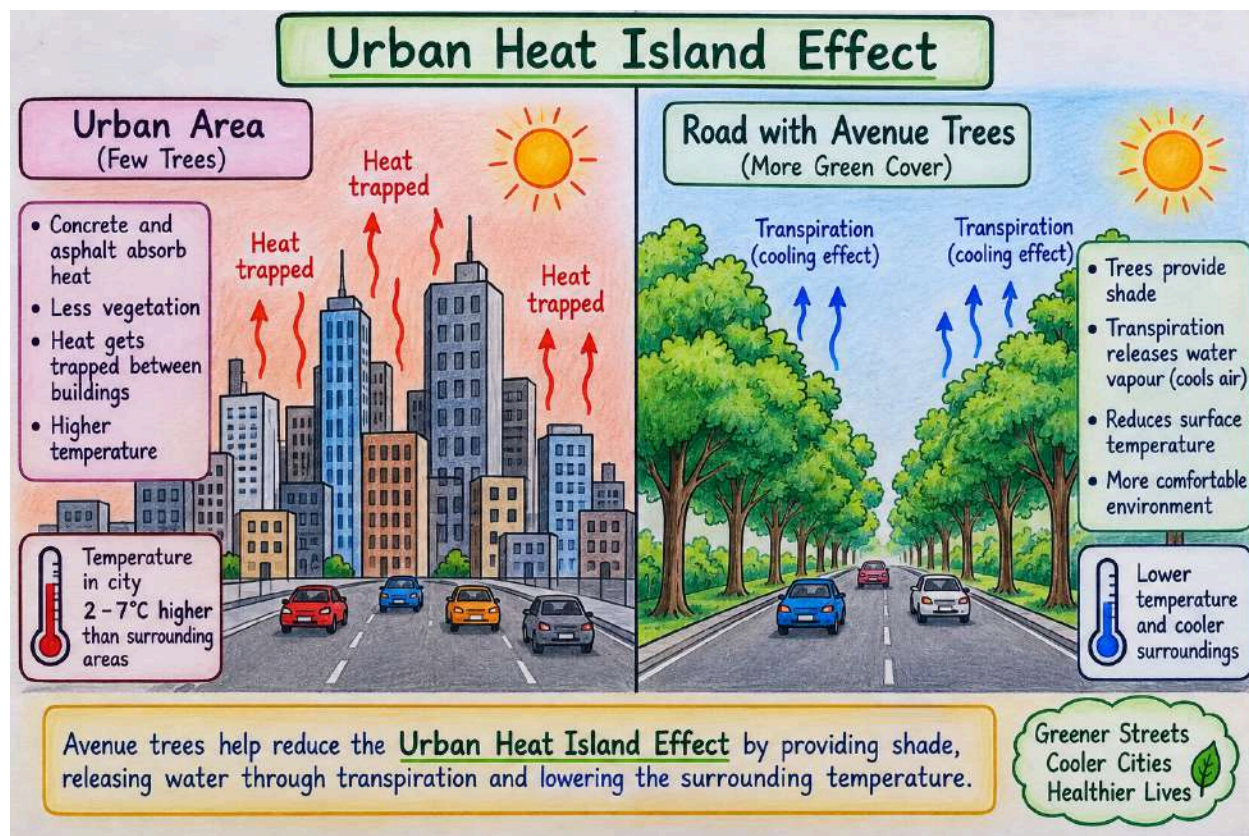
a. Introduction of the topic and need for the project Introduction to Arboriculture and

Avenue Trees: Avenue trees are trees planted in linear rows along the sides of roads, streets, highways, and boulevards. The practice of planting shade-giving trees along routes dates back to ancient Indian civilization, notably promoted by Emperor Ashoka. Today, in the context of modern biology and urban ecology, avenue trees are recognized as the "lungs" of urban and semi-urban environments.

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(Action: Draw a diagram showing the "Urban Heat Island Effect". Draw a city with tall buildings

showing red arrows (Heat trapped) and next to it, a road lined with green Avenue trees showing blue arrows (Cooling/Transpiration).)



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They are carefully selected based on their canopy size, root structure, foliage density, and resistance to urban pollution. A well-planned avenue plantation consists of a mix of native species that provide dense shade, medicinal benefits, and act as a refuge for local urban wildlife (avifauna and insects).

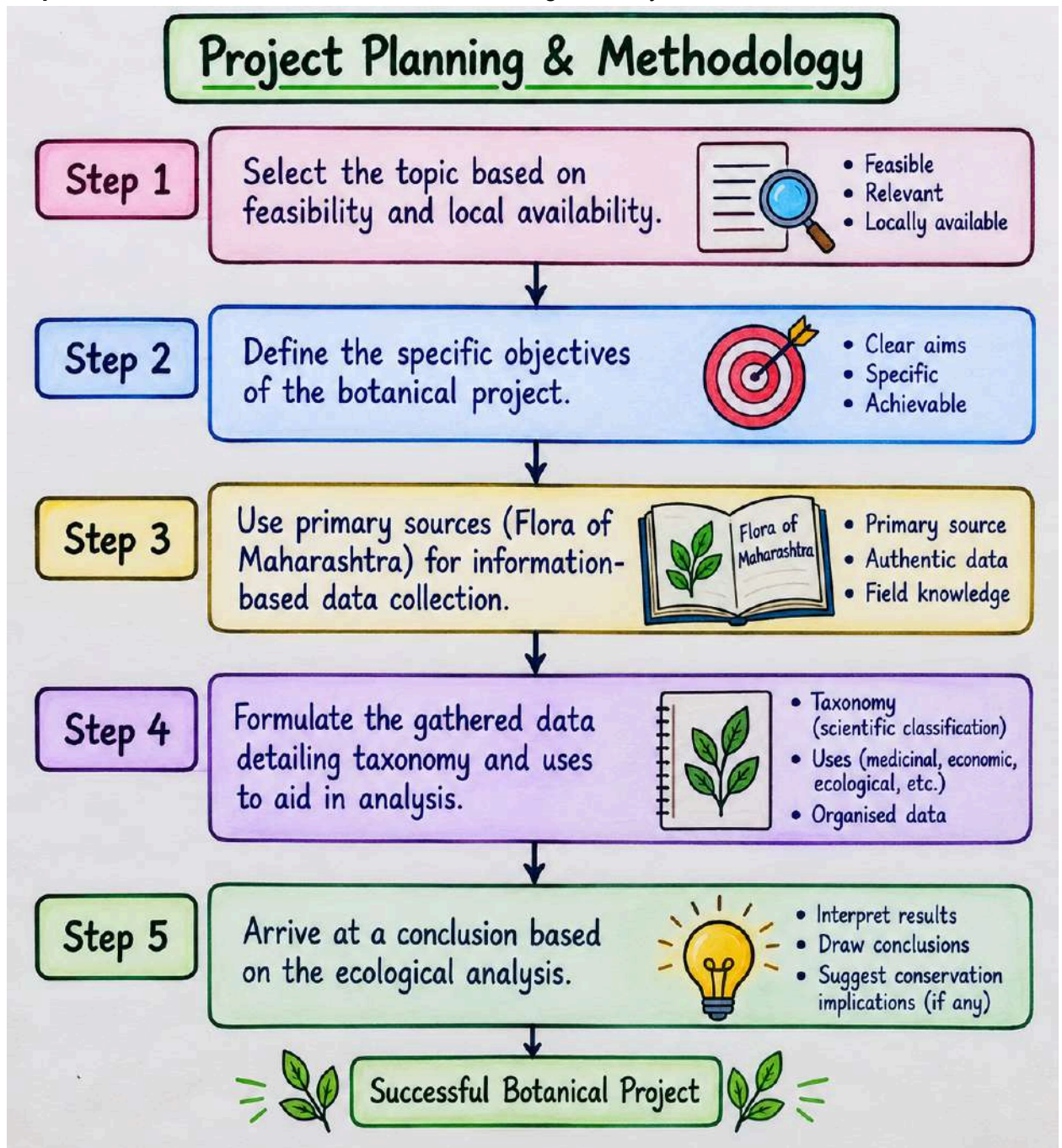
Need for the Project: With rapid urbanization, vehicular pollution, and the expansion of concrete infrastructure, cities are increasingly suffering from the "Urban Heat Island" effect and severe air quality degradation. This project is necessary to document the common avenue trees in our surroundings, understand their specific biological adaptations, and highlight their immense ecological importance. By studying these trees, we can promote better urban forestry practices and raise awareness about protecting urban green covers.

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(Action: Draw a neat flowchart titled "Project Planning & Methodology".)

- **Step 1:** Select the topic based on feasibility and local availability.
- **Step 2:** Define the specific objectives of the botanical project.
- **Step 3:** Use primary sources (Flora of Maharashtra) for information-based data collection.
- **Step 4:** Formulate the gathered data detailing taxonomy and uses to aid in analysis.

- **Step 5:** Arrive at a conclusion based on the ecological analysis.



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b. Description of method/s used As per the project guidelines, this is an **information-based project** focusing on primary sources.

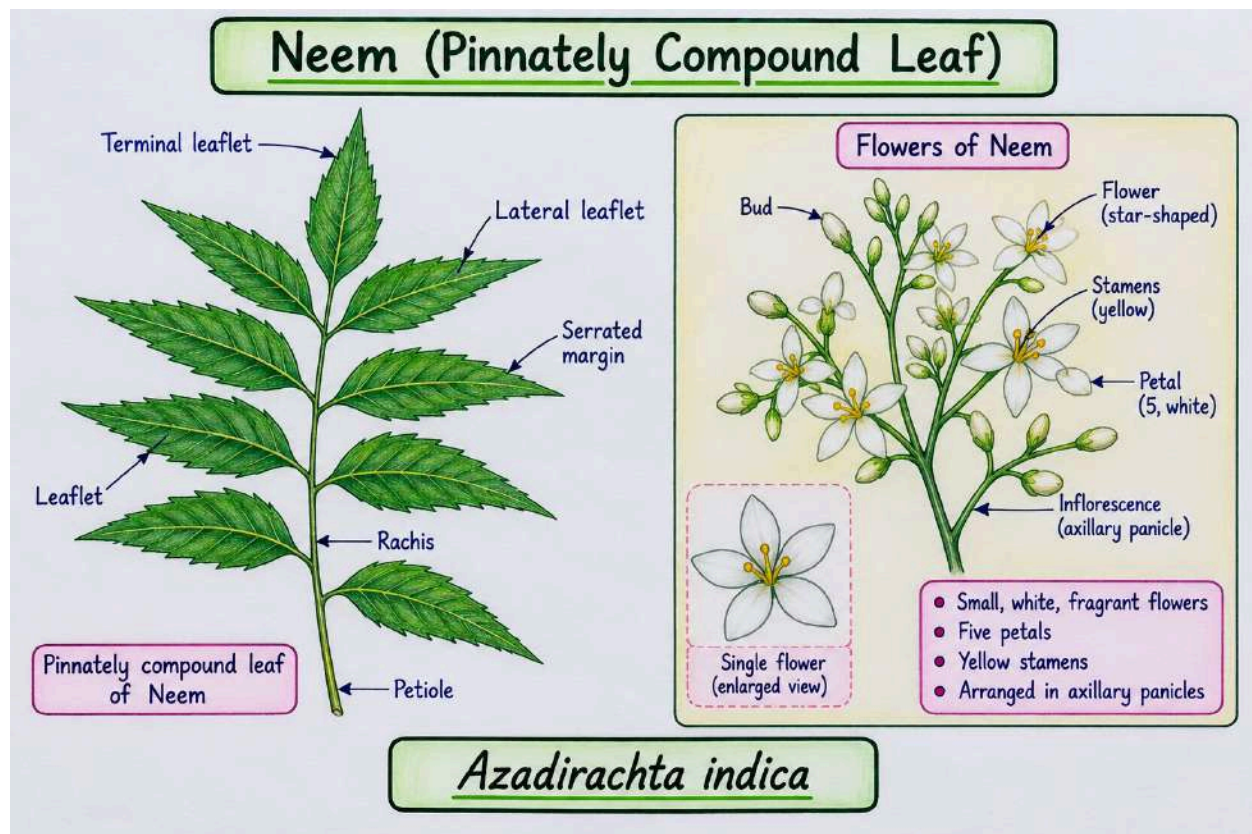
- **Primary Sources and Observation:** Data was meticulously collected through direct field observation of avenue trees in the local surrounding area. The field data was then cross-referenced with highly reliable botanical texts, primarily the "Flora of Maharashtra"

and the Botanical Survey of India (BSI) databases, to ensure accurate taxonomic classification.

- **Data Formulation:** Specific details regarding the morphological characteristics and localized distribution were gathered using various techniques so as to aid in analysis.
- **Data Organization:** The information was systematically categorized into specific tree species, detailing their scientific classification, morphological features, and their ecological and economic importance.

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(Action: Print and paste or draw a neat diagram of a Neem leaf (pinnately compound) and its small white flowers. Label it: *Azadirachta indica*).



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c. Data collected

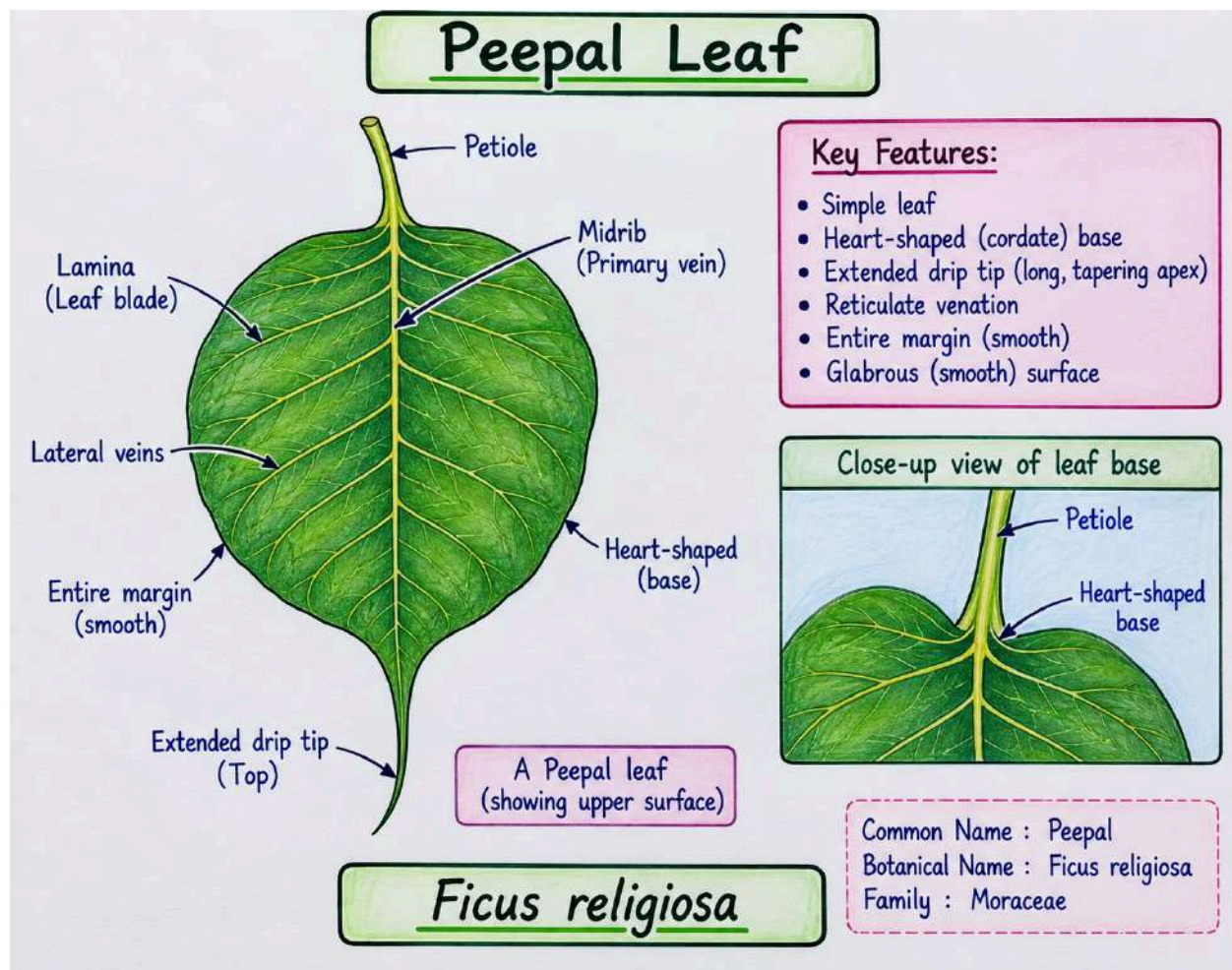
1. Neem (Indian Lilac)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Sapindales
 - Family: Meliaceae
 - Scientific Name: *Azadirachta indica*

- **Morphological Characteristics:** A fast-growing, evergreen tree with a broad canopy. Leaves are pinnately compound with serrated edges. It produces small, fragrant white flowers and smooth, olive-like drupe fruits.
- **Ecological Importance:** Extremely drought-resistant and tolerates high temperatures. Its dense foliage provides excellent shade and filters particulate matter from vehicular exhausts.
- **Economic/Medicinal Importance:** Known as the "village pharmacy." Every part of the tree has antibacterial and antifungal properties. Neem oil acts as a powerful natural biopesticide.

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(Action: Print and paste or draw a diagram of a Peepal leaf (heart-shaped with an extended drip tip). Label it: *Ficus religiosa*).



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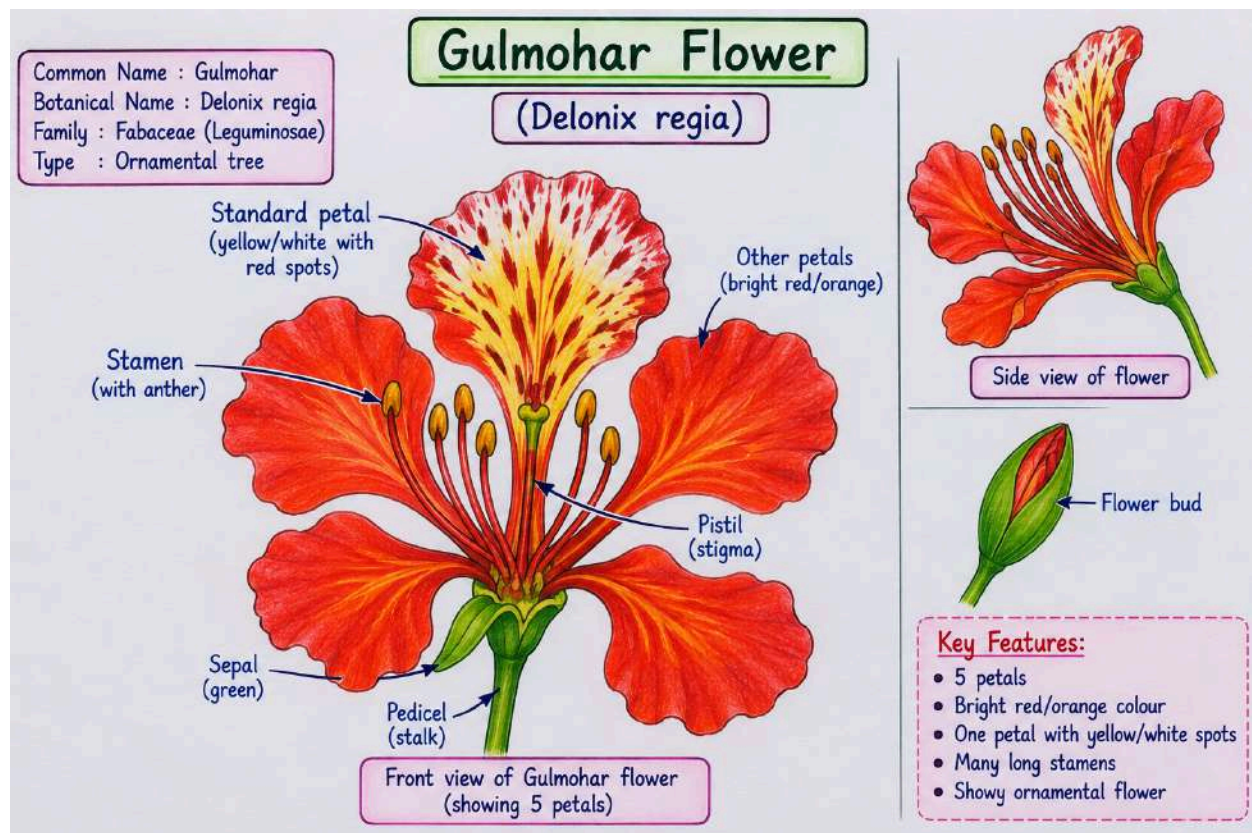
2. Peepal (Sacred Fig)

- **Taxonomic Classification:**

- Kingdom: Plantae
- Clade: Angiosperms
- Order: Rosales
- Family: Moraceae
- Scientific Name: *Ficus religiosa*
- **Morphological Characteristics:** A large dry-season deciduous tree. Leaves are distinctively heart-shaped (cordate) with an extended "drip tip." The tree relies on a specialized species of fig wasp for pollination.
- **Ecological Importance:** Peepal trees are considered keystone species in urban environments. They produce massive amounts of oxygen and their figs sustain a wide variety of urban birds (like hornbills and barbets), bats, and monkeys during dry seasons.
- **Economic/Medicinal Importance:** The bark is used in traditional Ayurvedic medicine to heal stomach ulcers and skin diseases. It holds immense cultural and religious significance, ensuring its protection.

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(Action: Print and paste or draw a diagram of the Gulmohar flower (5 bright red/orange petals, one with yellow/white spots). Label it: *Delonix regia*).



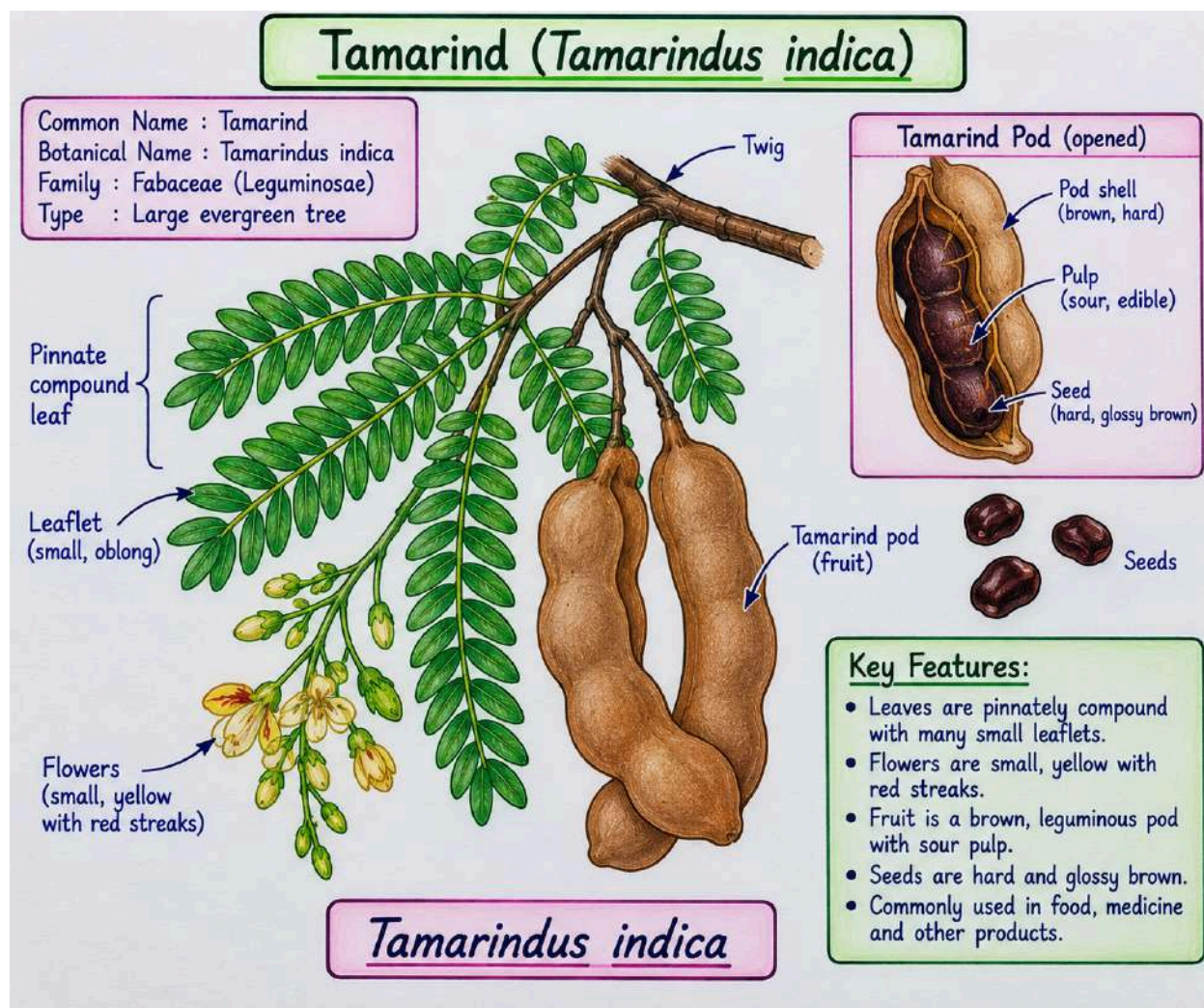
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3. Gulmohar (Royal Poinciana)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Fabales
 - Family: Fabaceae
 - Scientific Name: *Delonix regia*
- **Morphological Characteristics:** An exotic (non-native) but widely planted avenue tree. It has a distinctive umbrella-shaped canopy with feathery, bipinnately compound leaves. It produces flamboyant clusters of large red-orange flowers in the summer.
- **Ecological Importance:** Planted primarily for its rapid growth, extensive shade coverage, and highly aesthetic floral display, which beautifies urban landscapes.
- **Economic/Medicinal Importance:** While its timber is soft and of low commercial value, its deep root system helps in soil binding along road embankments, preventing soil erosion during heavy monsoons.

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(Action: Print and paste or draw a diagram of Tamarind pods and its small pinnate leaves. Label it: Tamarindus indica).



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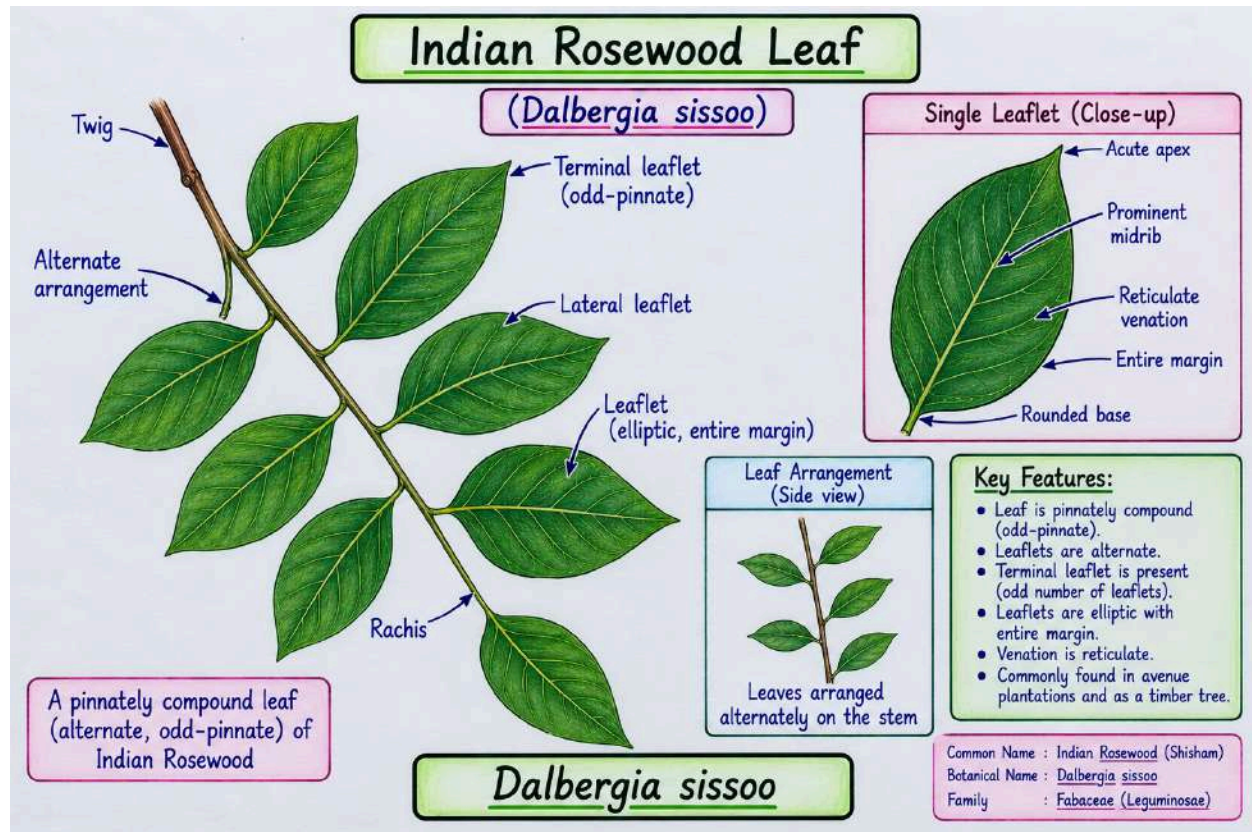
4. Tamarind (Chinch)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Fabales
 - Family: Fabaceae
 - Scientific Name: *Tamarindus indica*
- **Morphological Characteristics:** A massive, slow-growing, long-lived evergreen tree with dense, drooping foliage. It produces elongated, brown, pod-like fruits containing a highly acidic, edible pulp.
- **Ecological Importance:** Tamarind trees have an incredibly strong and deep root system, making them highly wind-resistant and perfect for highway avenues where cyclonic winds can uproot weaker trees.
- **Economic/Medicinal Importance:** Highly valuable economically. The fruit pulp is a staple ingredient in Indian cuisine. The timber is exceptionally hard and termite-resistant, used

for making specialized tools and furniture.

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(Action: Print and paste or draw a diagram of the Indian Rosewood leaf (alternate, odd-pinnate). Label it: *Dalbergia sissoo*).



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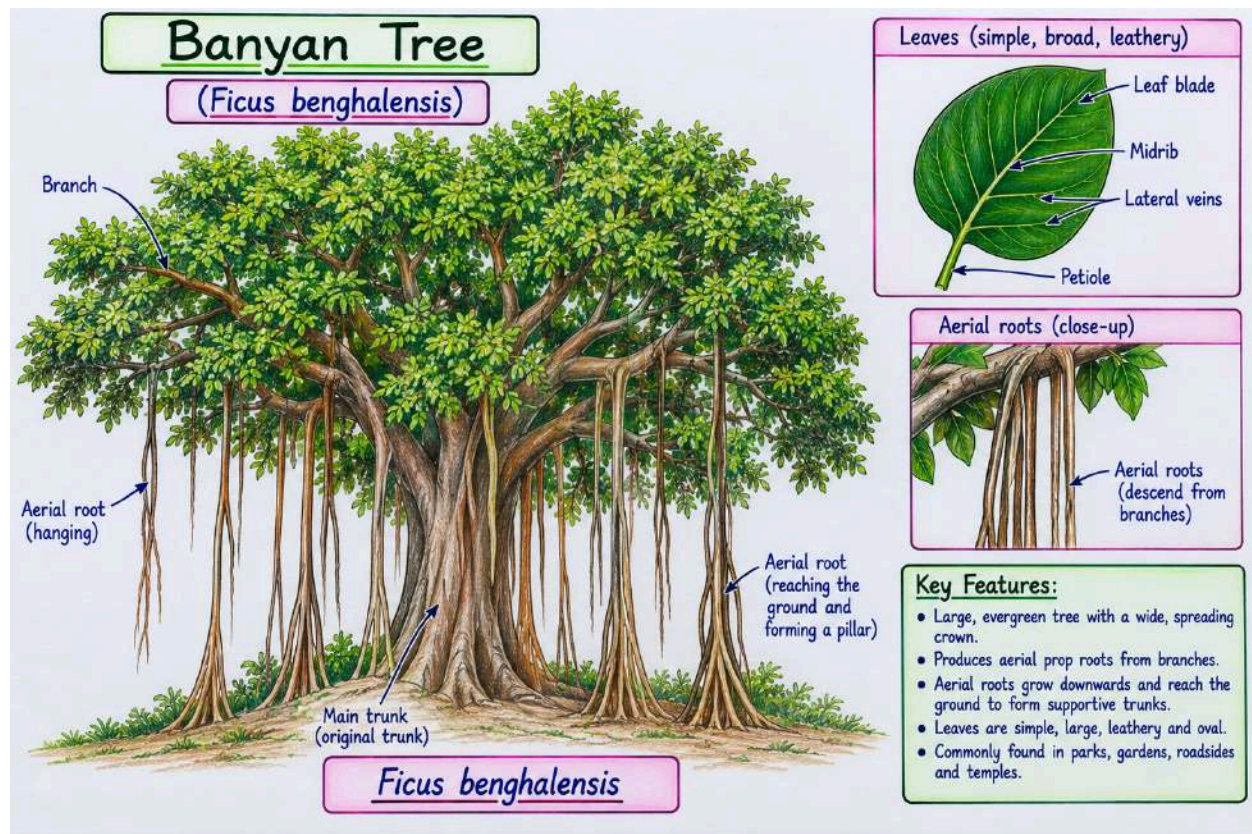
5. Shisham (Indian Rosewood)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Fabales
 - Family: Fabaceae
 - Scientific Name: *Dalbergia sissoo*
- **Morphological Characteristics:** A medium to large deciduous tree with a light crown. Leaves are leathery and pinnately compound. It produces small, pale yellow flowers.
- **Ecological Importance:** Shisham is a leguminous tree, meaning it has a symbiotic relationship with nitrogen-fixing bacteria in its root nodules. Planting it along avenues naturally enriches the nutrient-depleted soil found near roadsides.
- **Economic/Medicinal Importance:** It yields one of the finest premium timbers in the world (Rosewood), highly prized for musical instruments and luxury cabinetry due to its rich

color and durability.

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(Action: Print and paste or draw a diagram of the Banyan tree showing its aerial prop roots descending to the ground. Label it: *Ficus benghalensis*).



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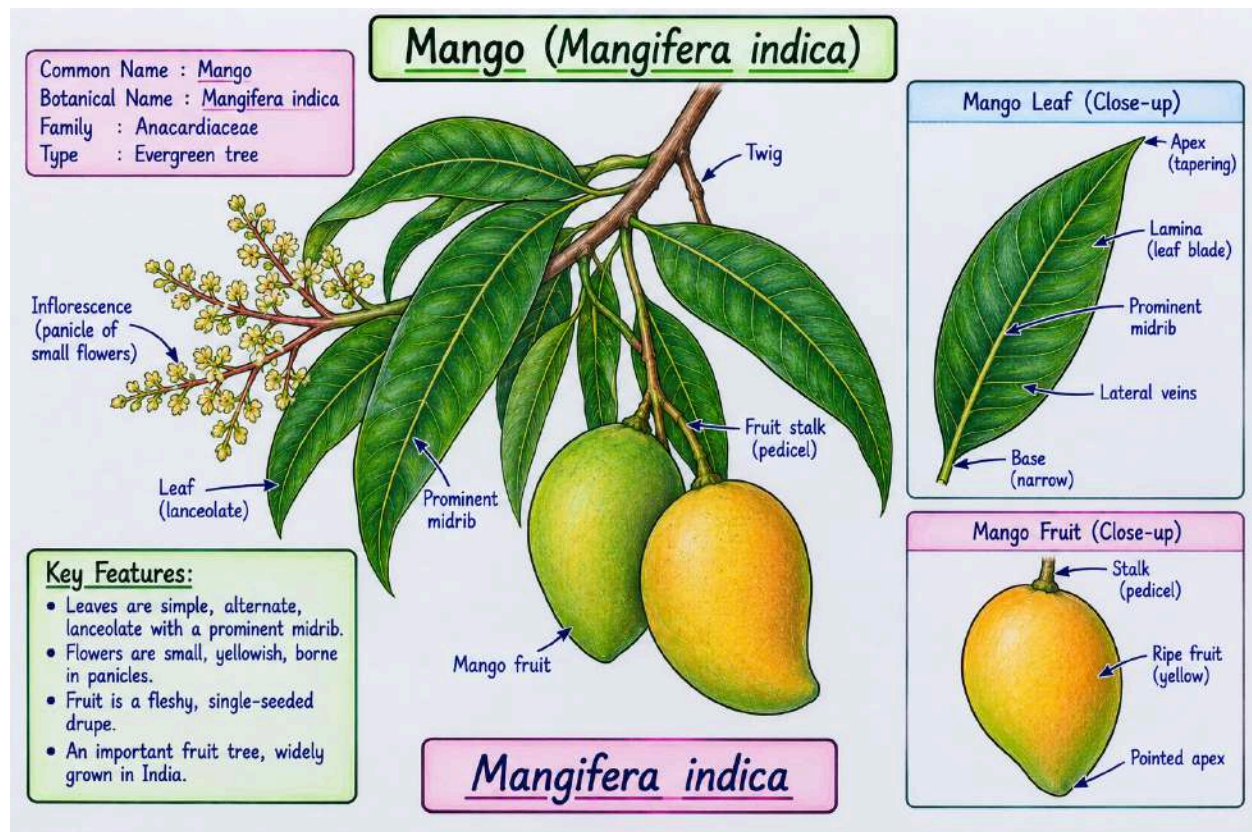
6. Banyan (Vad)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Rosales
 - Family: Moraceae
 - Scientific Name: *Ficus benghalensis*
- **Morphological Characteristics:** The National Tree of India. It is a massive evergreen tree characterized by its vast canopy and unique aerial prop roots that grow downward and anchor into the soil, eventually acting as accessory trunks.
- **Ecological Importance:** A single mature Banyan tree functions as an entire micro-ecosystem. Its dense canopy provides unmatched shade for commuters, while its figs support massive populations of birds and frugivorous mammals.
- **Economic/Medicinal Importance:** The latex (milky sap) is used in traditional medicine to

treat inflammation and skin ailments. Its massive structure acts as a natural windbreak and sound barrier on busy highways.

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(Action: Print and paste or draw a diagram of a Mango leaf and fruit. Label it: *Mangifera indica*).



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7. Mango Tree (Amba)

- **Taxonomic Classification:**
 - Kingdom: Plantae
 - Clade: Angiosperms
 - Order: Sapindales
 - Family: Anacardiaceae
 - Scientific Name: *Mangifera indica*
- **Morphological Characteristics:** A large, long-lived evergreen tree with a heavy, dome-shaped crown. Leaves are dark green, simple, and lanceolate. It bears fleshy drupe fruits.
- **Ecological Importance:** Widely planted along state highways for its dense, cooling shade. It acts as an excellent carbon sink, absorbing large quantities of CO₂ from vehicular emissions.
- **Economic/Medicinal Importance:** It is economically crucial for its highly demanded

seasonal fruit crop. The bark and leaves contain mangiferin, which possesses strong antioxidant and anti-inflammatory properties used in herbal remedies.

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Botanical Summary of Avenue Trees

Common Name	Scientific Name	Family	Primary Avenue Utility
Neem	<i>Azadirachta indica</i>	Meliaceae	Air purification & Shade
Peepal	<i>Ficus religiosa</i>	Moraceae	High Oxygen output
Gulmohar	<i>Delonix regia</i>	Fabaceae	Aesthetic beautification
Tamarind	<i>Tamarindus indica</i>	Fabaceae	Wind resistance & Shade
Shisham	<i>Dalbergia sissoo</i>	Fabaceae	Soil nitrogen fixation
Banyan	<i>Ficus benghalensis</i>	Moraceae	Massive shade & Sound barrier
Mango	<i>Mangifera indica</i>	Anacardiaceae	Carbon sink & Food yield

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d. Analysis of collected data

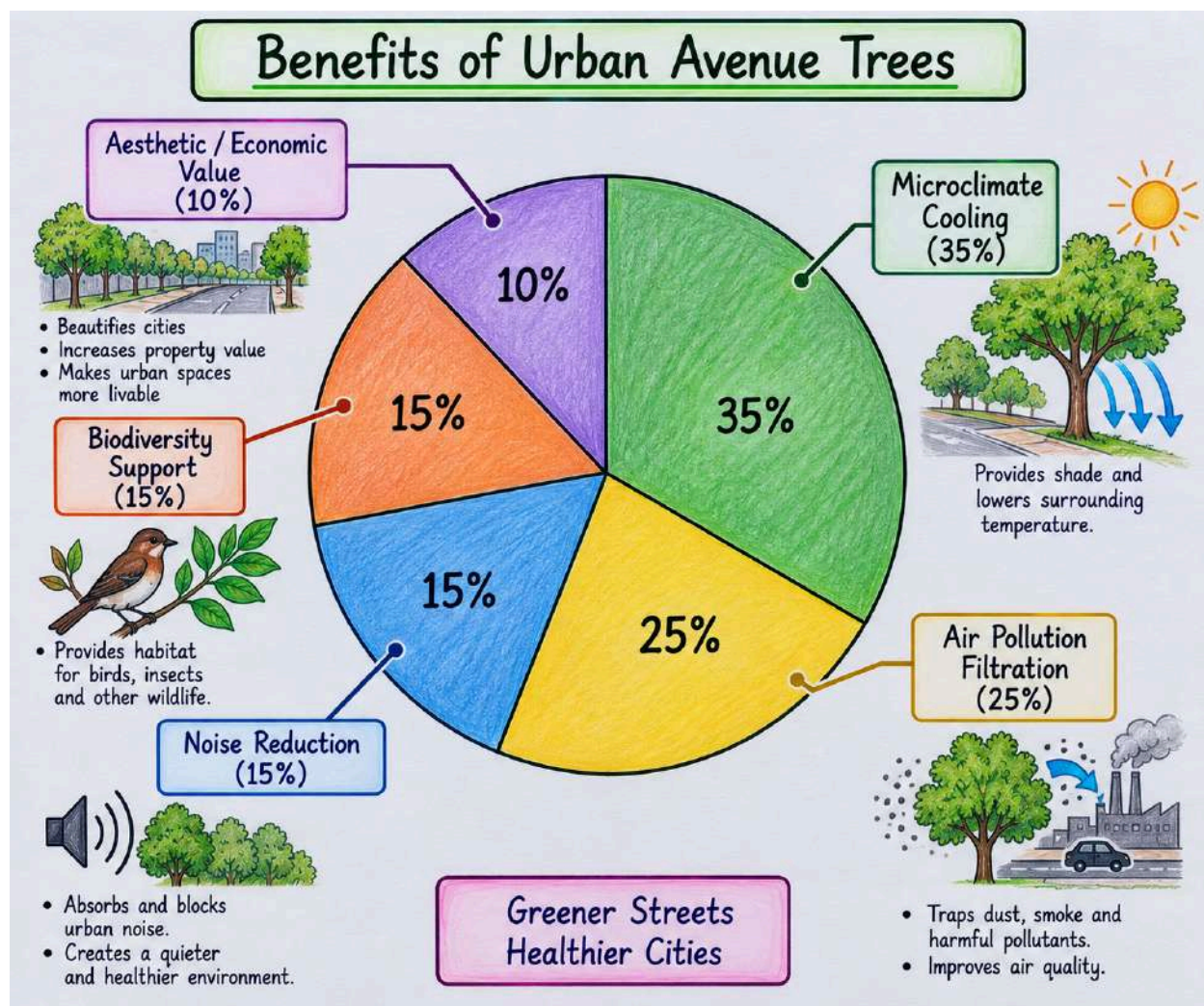
Through the formulation and study of the gathered botanical information, a clear pattern regarding the multi-faceted importance of avenue trees in urban and semi-urban landscapes emerges.

1. Environmental Filtration and Micro-Climate Regulation: The data definitively shows that avenue trees serve as biological filters. The dense canopies of trees like the Banyan and Neem physically trap airborne particulate matter (PM 2.5 and PM 10) generated by vehicular exhaust. Furthermore, through the process of evapotranspiration, these trees release moisture into the air, significantly reducing the localized ambient temperature of asphalt roads and combating the Urban Heat Island effect.

2. Acoustic and Wind Barriers: The analysis reveals that trees with massive trunks, dense foliage, and prop roots (like *Ficus benghalensis*) act as highly effective natural sound-absorbing barriers, reducing noise pollution from heavy highway traffic.

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(Action: Draw a Pie Chart titled "Benefits of Urban Avenue Trees": Microclimate Cooling (35%), Air Pollution Filtration (25%), Noise Reduction (15%), Biodiversity Support (15%), Aesthetic/Economic Value (10%).)



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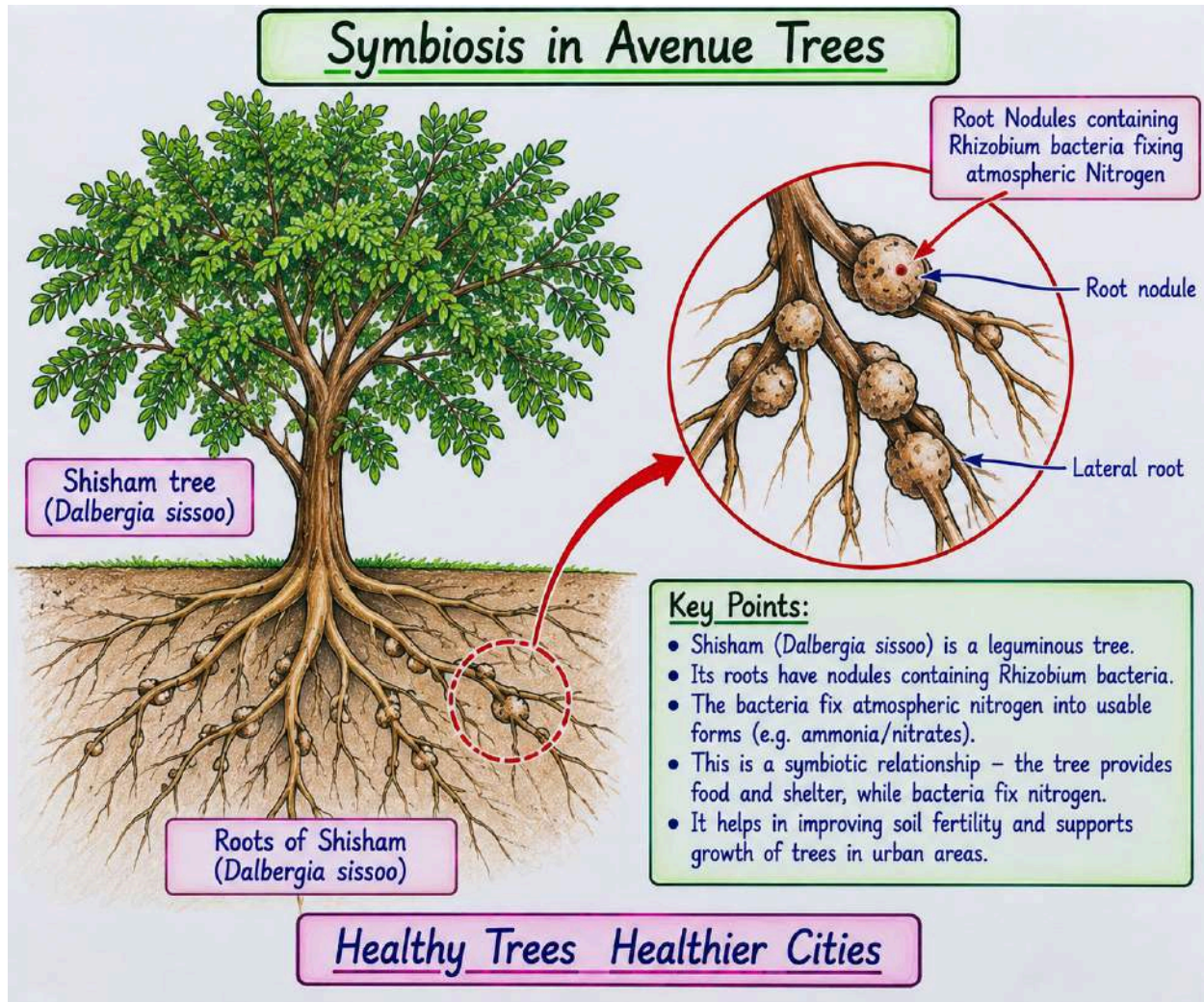
Additionally, deep-rooted species like the Tamarind provide crucial wind resistance, acting as windbreaks that protect adjacent agricultural lands and settlements from strong cyclonic winds.

3. Ecological Role and Biodiversity Support: The compiled data highlights that native avenue trees (especially the *Ficus* family, like Peepal and Banyan) function as keystone species. They act as vital green corridors for urban wildlife, providing nesting spaces, safety, and a year-round food supply (figs) for birds, insects, and small mammals that would otherwise be displaced by urbanization.

4. The Native vs. Exotic Dilemma: The data analysis also points to an important urban planning consideration. While exotic trees like the Gulmohar are frequently planted for their rapid growth and stunning floral aesthetics, they possess shallow root systems making them prone to uprooting during monsoons. In contrast, native species (Neem, Tamarind) offer superior longevity, better storm resistance, and higher ecological compatibility with local fauna.

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(Action: Draw a diagram titled "Symbiosis in Avenue Trees". Draw the roots of the Shisham tree (*Dalbergia sissoo*) showing small circles on the roots labeled "Root Nodules containing Rhizobium bacteria fixing atmospheric Nitrogen".)



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5. Soil Conservation and Enrichment: Certain avenue trees offer invisible, subterranean benefits. Leguminous trees like Shisham (*Dalbergia sissoo*) form symbiotic relationships with nitrogen-fixing bacteria in their root nodules. This biologically enriches the often poor, compacted soil found along roadsides, promoting the growth of secondary underbrush which prevents topsoil erosion during heavy rainfall.

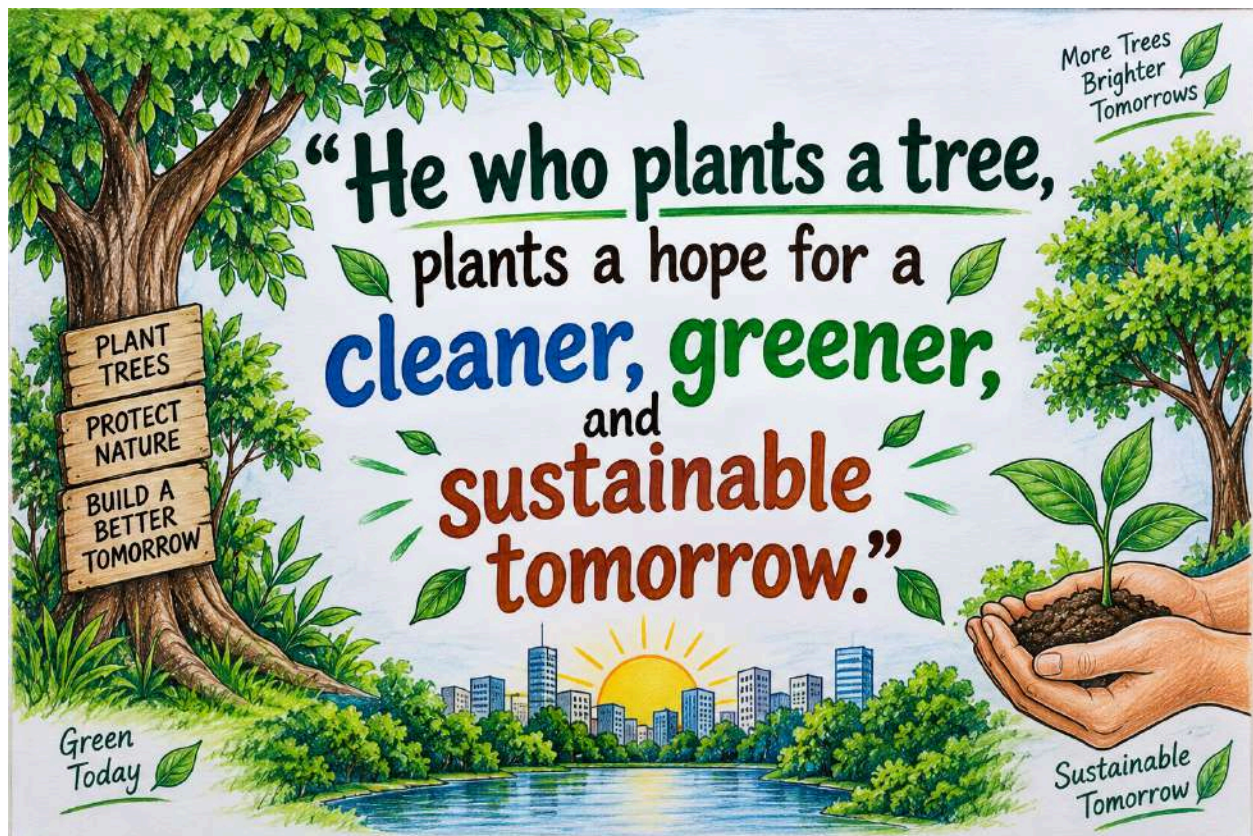
Conservation and Planting Strategies: Based on the analysis, future urban planning should prioritize planting native species with deep taproots and broad canopies. A mixed-plantation approach (polyculture) should be utilized along highways to prevent the spread of

species-specific plant diseases, ensuring a resilient urban green cover.

e. Conclusion: Arriving at a conclusion based on the analysis of data, it is evident that avenue trees are not mere decorative elements of urban infrastructure; they are indispensable biological assets.

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(Action: Write a bold, prominent quote in the center of the page: "He who plants a tree, plants a hope for a cleaner, greener, and sustainable tomorrow.").



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This project successfully fulfills its primary objective of identifying various avenue trees and documenting their importance. The data conclusively proves that these trees provide critical ecosystem services ranging from carbon sequestration and microclimate cooling to soil conservation and biodiversity support.

The presence of avenue trees directly correlates with the physical and mental well-being of urban populations by mitigating vehicular pollution and providing essential green spaces. Therefore, it is imperative that local municipalities, highway authorities, and citizens strictly protect existing mature trees from illegal felling and prioritize scientific arboriculture in all future infrastructure development. Ultimately, integrating native botanical diversity into our cities is the

only sustainable path forward.

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f. Bibliography- references for all the information sighted in the project

Textbooks and Academic Resources:

1. State Board Biology Textbook for Class 12, Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune. (Referenced for core concepts of Plant Morphology and Ecology).
2. NCERT Biology Class 11 and 12 Textbooks (Used for foundational taxonomic classifications and anatomical features).

Botanical & Scientific Reports:

1. "Flora of Maharashtra State" – Research publications and survey data by the Botanical Survey of India (BSI), Western Regional Centre, Pune.
2. "Trees of Delhi: A Field Guide" by Pradip Krishen (Used as a reference for urban arboriculture and species identification).
3. Reports on Urban Forestry and Air Pollution Mitigation – Published by the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India.

Online Scientific Databases:

1. eFlora of India Database (BSI) - Accessed for confirming the exact scientific nomenclature and morphological traits of the listed flora.
2. Discover Life (discoverlife.org) - Accessed for taxonomic verification.